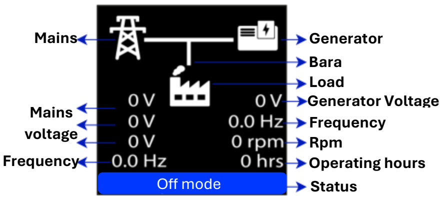
ENKO ELEKTRONİK

AMF-S GENERATOR CONTROL SYSTEM

MAIN MENU AND KEY IDENTIFICATION



- ››› START
- ››› STOP
- ››› ENTER AND RIGHT
- ››› EXIT, LEFT AND RESET
- ››› DOWN (ALARM AND FAULT) UP

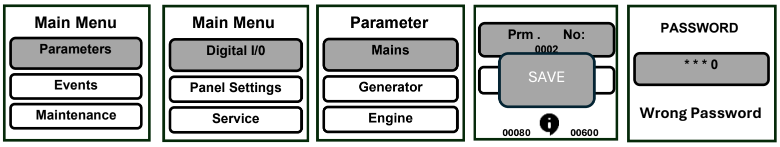
START BUTTON: Used to crank the generator in MSU Manual start Mode or AMF Test Mode. Also, while the generator is running in these modes, pressing the Start Button again will engage the generator load.

STOP BUTTON: When the Stop Button is pressed, if the generator has engaged the load, it will transition to cooling. If the generator has not engaged the load, it will directly enter the stopping state. Pressing this button switches the device to Closed Mode.

AUTO BUTTON: The Auto Button is used for mode transitions in the device during short presses on the monitoring pages. If an AMF Module is defined for the device, selections are made between Off Mode, Test Mode, and Auto Mode. If an MSU Module is defined, selections between Off mode and Manual mode are made. Additionally, this button is used as the Enter button during short presses on Menu pages.

EXIT BUTTON: The Exit Button is used to exit the menus entered. A short press of the Exit Button while on Monitoring Pages performs a reset operation.

MAIN MENU AND USER INTERFACE



Press the Enter key to access the main menu. Use the Up and Down buttons to change the selection. The key is used to exit and access the sub-menus.

PARAMETER MODIFICATION

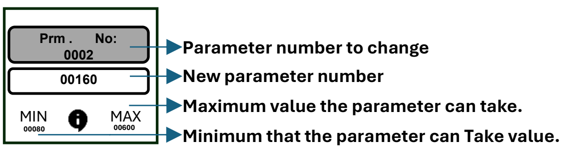
Press the Enter key from the Main Menu and select the Parameters tab.

Enter the Enter Parameter Number tab with the Enter key.

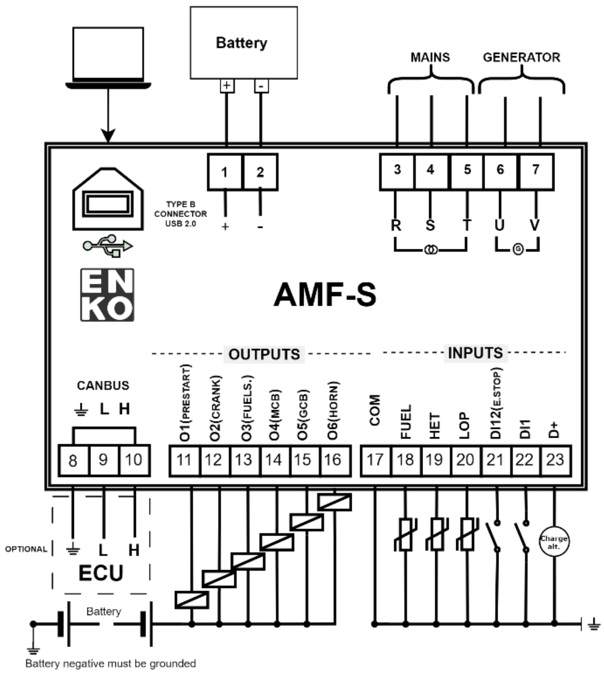
Use the keys for up and down to enter the values you want to change.

Use the keys for right and left digits to enter and confirm with the Enter key.

To exit from the menu without changing parameters, press and hold the Exit key.

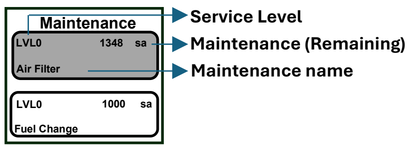


AMF-S CONNECTION DIAGRAM



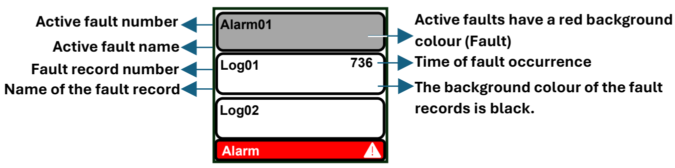
SERVICE TIME RESET

The service time you wish to reset is selected with the Up and Down buttons. Then, press and hold the Exit button for 5 seconds.



ACTIVE FAULTS AND FAULT RECORDS

Active faults are listed on this page with a red background color. The fault record list begins right below the list of active faults. It can be accessed by pressing the Down button on the main page.



PARAMETERS

	P.	PARAMETER DESCRIPTION	UNI.	L	MİN.	MAX.	DEF
GRID	101	Closed Mode Selection		2	0	1	0
	102	High Voltage Alarm Level	%	1	101	150	115
	103	Low Voltage Alarm Level	%	1	50	99	85
	104	High Frequency Alarm Level	%	1	101	150	104
	105	Low Frequency Alarm Level	%	1	50	99	96
	110	Phase Sequence Control Action		1	0	1	1
	111	Connection Type		2	0	1	1
ENGINE	301	Generator Connection Type		2	0	1	0
	302	Engine Type		3	0	1	0
	303	Module Function		3	0	1	0
	304	Mode Function		1	0	3	0
	305	Pre-start Action		2	0	2	0
	306	Starter Cut-off Level from Generator Frequency	Hz	3	150	750	15.0
	307	Starter Cut-off Level from Generator Speed	rpm	3	500	6000	500
	308	Starter Cut-off Level from Generator Voltage	V	3	60	500	300
	309	Starter Cut-off Level from Charge Alternator Voltage	V	3	60	300	60
	310	Starter Cut-off Level from Oil Pressure	bar	3	10	100	3
	311	Maximum Number of Start Attempts		2	1	10	3
	312	Intermittent Horn Output		1	0	1	0
	313	Oil Pressure Unit	bar	1	0	1	0
	314	Low Oil Pressure Alarm Action		3	0	4	4
	315	Low Oil Pressure Alarm Level	bar	2	5	95	1
	316	Oil Pressure Sensor Open Circuit Action		2	0	4	4
	317	Temperature Unit	°C	1	0	1	0
	318	High Coolant Temperature Alarm Action		3	0	4	4
	319	High Coolant Temperature Alarm Level	°C	2	5	150	110
	320	Coolant Temperature Sensor Open Circuit Action		2	0	4	4
	321	Low Fuel Level Alarm Action		3	0	4	4
	322	Low Fuel Level Alarm Level	%	2	0	45	5
	323	Fuel Level Sensor Open Circuit Action		2	0	4	4
	324	Fuel Pump Lower Limit	%	2	0	90	20
	325	Fuel Pump Upper Limit	%	2	5	95	80
	326	Cooling Fan Low Limit	°C	3	0	240	65
	327	Cooling Fan High Limit	°C	3	5	245	100
	328	Nominal Battery Voltage	V	2	100	260	130
	329	Battery High Voltage Alarm Action		3	0	4	4
	330	Battery High Voltage Alarm Level	%	2	101	125	125
	331	Battery Low Voltage Alarm Action		3	0	4	4
	332	Battery Low Voltage Alarm Level	%	2	75	99	75
	334	Charge Alternator High Voltage Alarm Action		3	0	4	4
	335	Charge Alternator High Voltage Alarm Level	%	2	101	125	125
	336	Charge Alternator Low Voltage Alarm Action		3	0	4	0
	337	Charge Alternator Low Voltage Alarm Level	%	2	75	99	75
	343	Maintenance Alarm (Oil) Action		2	0	4	2
	344	Maintenance Alarm (Air) Action		2	0	4	2
	345	Maintenance Alarm (Fuel) Action		2	0	4	2
	346	General Maintenance Alarm Action		2	0	4	2

	P.	PARAMETER DESCRIPTION	UNI.	L	MİN.	MAX.	DEF
GENERAL	2	Factory Password		3	0	9999	1923
	3	Service Password		2	0	9999	1922
	4	User Password		1	0	9999	1934
	5	Parameter Recording		1	0	2	0
	6	LANGUAGE		1	0	1	0
	7	Return to Factory Settings		3	0	2	0
	8	Log Clearing		3	0	1	0
	9	Engine Hour Setting		3	0	32000	0
	10	Menu Timeout	min	3	1	30	5
	11	Exit Menu		1	0	1	0
GENERATOR	202	Alternator Pole Number		1	0	7	1
	203	Nominal Voltage	V	2	85	240	220
	205	High Voltage Alarm Action		3	0	4	4
	206	High Voltage Alarm Level	%	2	101	150	115
	207	Low Voltage Alarm Action		3	0	4	4
	208	Low Voltage Alarm Level	%	2	50	99	85
	209	Nominal Frequency	Hz	2	300	600	50.0
	211	High Frequency Alarm Action		3	0	4	4
	212	High Frequency Alarm Level	%	2	101	130	106
	213	Low Frequency Alarm Action		3	0	4	4
	214	Low Frequency Alarm Level	%	2	50	99	94
ENGINE TIMER	601	Starting Delay	s	1	0	6000	50
	602	Network Stabilization Duration	s	1	0	18000	200
	603	Pre-start Time	s	2	0	6000	20
	604	Maximum Starting Time	s	3	0	600	50
	605	Start Waiting Time	s	3	50	990	100
	606	Fault Control Delay	s	3	0	1000	100
	607	Choke Time	s	1	0	600	20
	608	Oil Pressure Switch Start Cut-off Time	s	3	0	50	0
	611	Stop Solenoid Timer		3	0	1200	200
	612	Engine Warming Time	s	3	0	3600	0
	613	Cooling Duration	s	3	0	18000	300
	615	Transfer Time	s	3	0	6000	7
	616	Horn Duration	s	2	10	900	30
	619	Low Oil Pressure Alarm Delay	s	3	1	600	30
	620	Oil Pressure Sensor Open Circuit Delay	s	2	1	600	30
	621	High Coolant Temperature Alarm Delay	s	3	1	600	50
	622	Coolant Temperature Sensor Open Circuit Delay	s	2	1	600	50
	623	Low Fuel Level Alarm Delay	s	3	1	600	10
	624	Fuel Level Sensor Open Circuit Delay	s	2	1	600	10
	631	Maintenance Alarm (Oil) Timer	h	3	200	10000	1000
	632	Maintenance Alarm (Air) Timer	h	3	200	10000	1000
	633	Maintenance Alarm (Fuel) Timer	h	3	200	10000	1000
	634	General Maintenance Alarm Timer	h	3	200	10000	1000
	635	Service Time Renewal		2	0	4	0

	Output 1	Output 2	Output 3	Output 4	Output 5	Output 6
Output Function	1101	1103	1105	1107	1109	1111
Default Value	10	1	2	8	6	4
Output Delay	701	702	703	704	705	706
Default Value	0	0	0	0	0	0
Output Contact Type	1102	1104	1106	1108	1110	1112
Default Value	NO	NO	NO	NO	NO	NO
0: Output Not Active	1: Start Output	2: Fuel Solenoid	3: Stop Solenoid	4: Horn Output		
6: Generator Contactor	8: Network Contactor	9: Load Ready Output	10: Pre-start	11: Choke Output		
14: Cooling Fan	15: Fuel Pump	22: General Alarm	49: Low Fuel Level Alarm	70: Oil Change Maintenance Alarm		
71: Air Filter Change Alarm	72: Fuel Change Maintenance Alarm	73: General Maintenance Alarm	75: Output in Automatic Mode	76: Output in Manual Mode		
77: Audible Warning Before Operation						
	Input 1	Input 2	Input 3	Input 4	Input 5	
Input Function	1201	1205	1304	1404	1504	
Default Value	0	1	0	0	0	
Input Delay	701	702	703	704	705	
Default Value	0	0	0	0	0	
Input Contact Type	1102	1104	1106	1108	1110	
Default Value	NC	NC	NC	NC	NC	
0 : Input Inactive	1 : Emergency Stop	2 : Remote Start/Stop	3 : Remote Operation/Load	4 : Panel Lock		
9 : Start Button Simulation	13 : Stop Button Simulation	16 : Alarm Disable	17 : Alarm Reset	18 : User Defined		

- **NOTE: If the menu exit time has elapsed since the last entered password, the password screen is displayed. It is requested to enter the password again.**

- **NOTE: Errors and alarms received while in the main menu are reset, but they are not removed from the main screen as information. When the main screen is reached, reset button is pressed again.**

- **NOTE: In test mode, when the start button is pressed for the first time, the genset is started, but if it is pressed for the second time, the genset feeds the load.**

- **NOTE: In the event of a fault or warning, the warning symbol is displayed in the lower right corner of the screen and the status bar flashes red and gray depending on the level of the fault.**